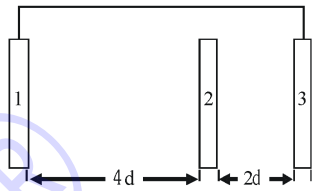
SECTION-I

Single Correct Answer Type **8 Q. [3 M (–1)]**

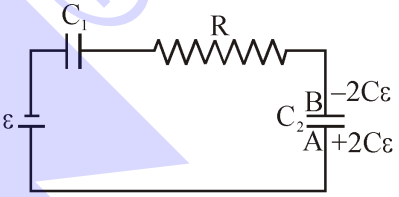
1. Three identical metal plates of large area A numbered 1,2,3 are arranged as shown in the diagram. The plates 1 and 3 are connected by a conducting wire. The charge given to plate 2 is 2q and no other plate is given any charge. Find the potential difference between the plate 2 and 3.

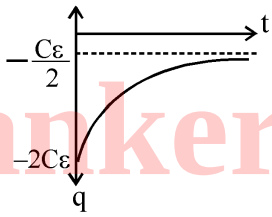


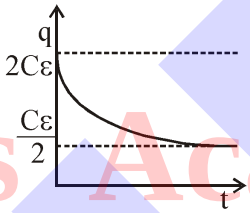
(A) $\frac{4}{3}\frac{q_0 d}{A\epsilon_0}$ (B) $\frac{8}{3}\frac{q_0 d}{A\epsilon_0}$

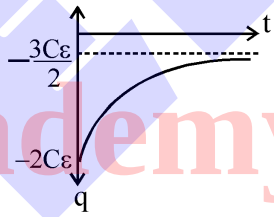
(C) $\frac{12}{3}\frac{q_0 d}{A\epsilon_0}$ (D) $\frac{2}{3}\frac{q_0 d}{A\epsilon_0}$

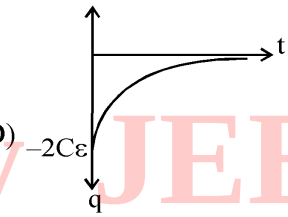
2. The capacitance of C_1 and C_2 shown in the diagram is C. C_1 is initially uncharged and C_2 is given a charge 2Cε as shown. Which of the following graph represents charge on plate B of capacitor C_2 as a function of time ?



(A) 

(B) 

(C) 

(D) 

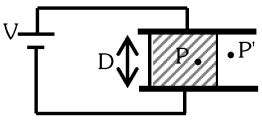
3. A parallel plate capacitor, partially filled with a dielectric slab of dielectric constant K, is connected with a cell of emf V volt, as shown in the figure. Separation between the plates is D. Then

(A) electric field at point P is less than that at point P′.

(B) electric field at point P′ is less than that at point P.

(C) electric fields at points P and P′ are equal

(D) electric field at point P is $E = \frac{V}{KD}$.



4. A metal ball of radius R is surrounded by a homogeneous medium of dielectric constant K and resistivity ρ. The ball is given a charge Q at t = 0, which starts leaking through the medium towards infinity. After how much time will the charge on the ball become Q/2?

(A) $[2K\rho\varepsilon_0].ln2$ (B) $[K\rho\varepsilon_0].ln2$ (C) $[K\rho\varepsilon_0/2].ln2$ (D) None

5. A parallel plate capacitor is filled with 3 dielectric materials of same thickness, as shown in the sketch. The dielectric constants are such that $\kappa_3 > \kappa_2 > \kappa_1$. Let the magnitudes of the electric field in and potential drops across each dielectric be E_3, E_2, E_1, ΔV_3, ΔV_2, and ΔV_1, respectively. Which one of the following statements is true?



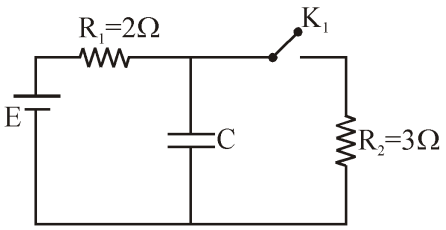
(A) $E_3 < E_2 < E_1$ and $\Delta V_3 < \Delta V_2 < \Delta V_1$ (B) $E_3 > E_2 > E_1$ and $\Delta V_3 > \Delta V_2 > \Delta V_1$

(C) $E_3 < E_2 < E_1$ and $\Delta V_3 > \Delta V_2 > \Delta V_1$ (D) $E_3 > E_2 > E_1$ and $\Delta V_3 < \Delta V_2 < \Delta V_1$

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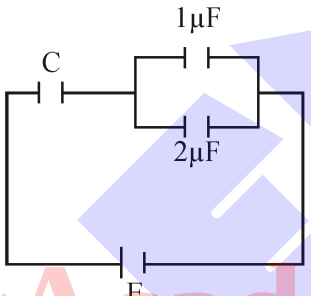
6. A dielectric slab is introduced between the plates of a capacitor. If the charge on the capacitor before the slab is introduced is q and the magnitude of the induced charge on the dielectric surface is q', then
(A) $q' < q$ (always) (B) $q' > q$ (always) (C) $q' = q$ (always) (D) $q' = 0$

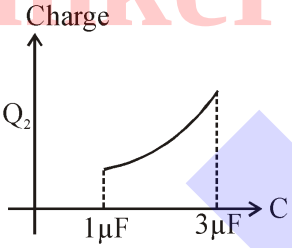
7. In the circuit shown in the figure K_1 is open. The charge on capacitor C in steady state is q_1. Now key is closed and at steady state charge on C is q_2. The ratio of charges q_1/q_2 is :

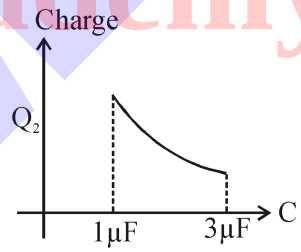


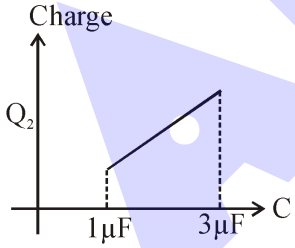
(A) 5/3 (B) 3/5 (C) 1 (D) 2/3

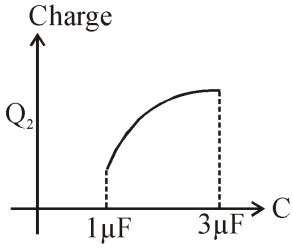
8. In the given circuit, charge Q_2 on the 2μF capacitor changes as C is varied from 1μF to 3μF. Q_2 as a function of 'C' is given properly by : (figures are drawn schematically and are not to scale) :-



(A)


(B)


(C)


(D)


Multiple Correct Answer Type

7 Q. [4 M (–1)]

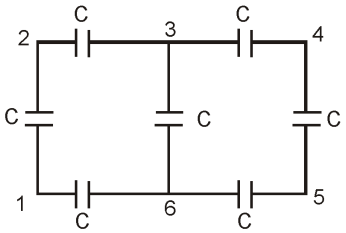
9. For the given circuit, select the correct alternative(s)

(A) The equivalent capacitance between points 1 & 2 is $\frac{15C}{11}$

(B) The equivalent capacitance between points 3 & 6 is $\frac{5C}{3}$

(C) The equivalent capacitance between points 1 & 3 is $\frac{15C}{14}$

(D) The equivalent capacitance between points 3 & 5 is $\frac{14C}{15}$



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10. A parallel plate air capacitor is connected to a battery. If plates of the capacitor are slowly pulled apart, then which of the following statements is/are **INCORRECT** ?

(A) Strength of electric field inside the capacitor remains unchanged, if battery is disconnected before pulling the plates.

(B) During the process, negative work is done by an external force applied to pull the plates whether battery is disconnected or it remains connected.

(C) Potential energy in the capacitor decreases if the battery remains connected during pulling plates apart.

(D) Potential energy in the capacitor decreases if the battery is disconnected before pulling plates apart.

11. We have a parallel plate capacitor made of two conducting, circular sheets of radius 2R each. They are kept at a small separation d between them. Suppose, we insert a circular shape dielectric of radius R and thickness d, between the gap with its centre coinciding with the plate centres the dielectric constant is k. The correct statement is/are :

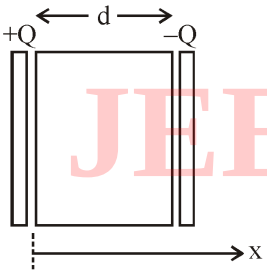
(A) Capacitance of the capacitor is $\frac{(k+3)\epsilon_0 \pi R^2}{d}$

(B) Capacitance of the capacitor is $\frac{k\epsilon_0 \pi(4R^2)}{d}$

(C) Ratio of electric field in the region with dielectric & without dielectric is 1

(D) Ratio of electric field in the region with dielectric & without dielectric is 1 : k

12. A dielectric slab is filled in between the plates of parallel plate charged capacitor. Di-electric constant of the capacitor varies with x as $k = \frac{x}{d}k_0$.



Capacitor is not connected with the battery. Then choose the correct statement(s) :-

(A) Net electric field inside the di-electric is uniform throughout the capacitor.

(B) Net electric field inside the dielectric decreases on moving along x-direction inside the capacitor.

(C) Bound charge density inside the di-electric is zero.

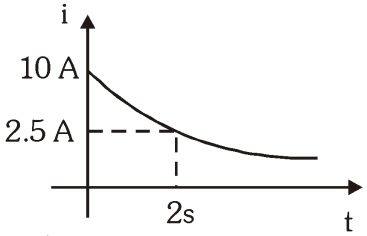
(D) Bound charge density inside the dielectric is non-uniform.

13. The figure, a graph of the current in a discharging circuit of a capacitor through a resistor of reistance 10Ω.

(A) The initial potential difference across the capactitor is 100 volt.

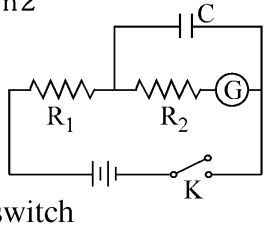
(B) The capacitance of the capacitor is $\frac{1}{10\ell n2}F$.

(C) The total heat produced in the circuit will be $\frac{500}{\ell n2}$ joules.



(D) The thermal power in the resistor will decrease with a time constant $\frac{1}{2\ell n2}$ second.

14. In the connection shown in the figure the switch K is open and the capacitor is uncharged. Then we close the switch and let the capacitor charge upto the maximum and open the switch again. The values indicated by the galvanometer.



(A) increases after closing the switch and then decreases after opening the switch

(B) direction of current is same through the galvanometer before and after opening the switch again

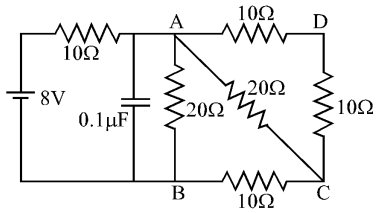
(C) after opening the switch again energy dissipation in R_1 is zero

(D) after opening the switch again energy dissipation in R_2 is zero

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15. A capacitor of capacitance 0.1 μF is connected to a battery of emf 8V as shown in the fig. Under steady state condition



(A) charge on the capacitor is 0.4 μC.

(B) charge on the capacitor is 0.2 μC

(C) current in the resistor between point A and B is 0.2 A

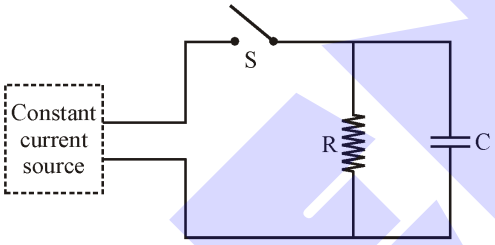
(D) current in the resistor between point A and B is 0.4 A

Linked Comprehension Type (2 Para × 2Q.) [3 M (-1)]

(Single Correct Answer Type)

Paragraph for Question Nos. 16 & 17

In the circuit shown in figure, switch S is closed at time t = 0. Thereafter the constant current source by varying its emf, maintains a constant current I out of its upper terminal.



16. Derive an expression for the charge on the capacitor as a function of time :-

(A) $q = iRC\left(e^{-\frac{t}{RC}}\right)$ (B) $q = iRC\left(1-e^{-\frac{t}{RC}}\right)$ (C) $q = it\left(1-e^{-\frac{t}{RC}}\right)$ (D) $q = \frac{i}{RC}\left(1-e^{-\frac{t}{RC}}\right)$

17. Find the time at which the rate of energy storage on capacitor is equal to the rate of heat dissipation through resistor :-

(A) $t = \frac{RC\,\ell n\,2}{2}$ (B) $t = 2RC\,\ell n\,2$ (C) $t = RC\,\ell n\,2$ (D) Not possible

Paragraph for Question Nos. 18 & 19

Consider a simple RC circuit as shown in figure 1.

Process 1 : In the circuit the switch S is closed at t = 0 and the capacitor is fully charged to voltage V_0 (i.e., charging continues for time T >> RC). In the process some dissipation (E_D) occurs across the resistance R. The amount of energy finally stored in the fully charged capacitor is E_C.

Process 2 : In a different process the voltage is first set to $\frac{V_0}{3}$ and maintained for a charging time T >> RC. Then the voltage is raised to $\frac{2V_0}{3}$ without discharging the capacitor and again maintained for a time T >> RC. The process is repeated one more time by raising the voltage to V_0 and the capacitor is charged to the same final voltage V_0 as in Process 1.

These two processes are depicted in Figure 2.

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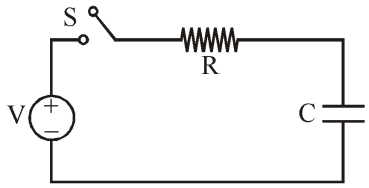


Figure 1

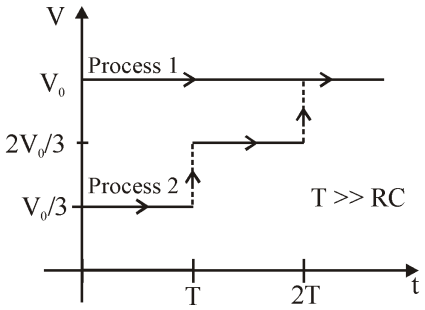


Figure 2

18. In Process 1, the energy stored in the capacitor E_C and heat dissipated across resistance E_D are related by:-

(A) $E_C = E_D$ (B) $E_C = 2E_D$ (C) $E_C = \frac{1}{2}E_D$ (D) $E_C = E_D \ln 2$

19. In Process 2, total energy dissipated across the resistance E_D is :-

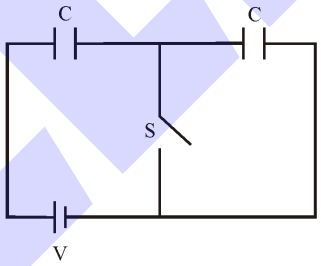
(A) $E_D = \frac{1}{3}\left(\frac{1}{2}CV_0^2\right)$ (B) $E_D = 3\left(\frac{1}{2}CV_0^2\right)$ (C) $E_D = \frac{1}{2}CV_0^2$ (D) $E_D = 3CV_0^2$

SECTION-II

Numerical Answer Type Question **1Q.[3M(0)]**

(upto second decimal place)

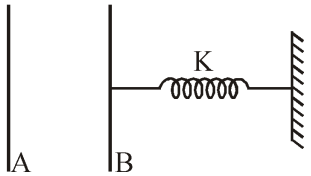
20. In the given circuit switch s is open initially. If C = 2μF and v = 3 volt. Then find heat produced (in μJ) in the circuit after closing the switch.



SECTION-III

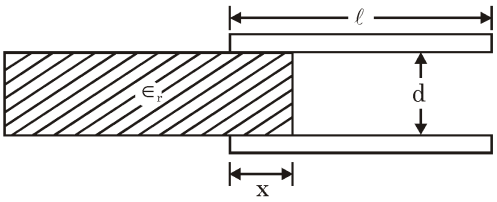
Numerical Grid Type (Ranging from 0 to 9) **2 Q. [4 M (0)]**

21. Initially the plates of the parallel plate capacitor shown in the figure are A and B containing charge +2μC and –2μC respectively. Negatively charged plate B is connected to a spring of spring constant 5000N/m at one end and the other end of the spring is connected to a rigid support. Now the plate A is fixed and the system is released from rest. The maximum elongation produced in the spring is $\frac{\alpha \times 10^{-12}}{\epsilon_0}$ m. The area of the plates is 2 cm^2, the medium between the plates is air and Electric permittivity of air is ϵ_0 (in SI). Then the value of α.



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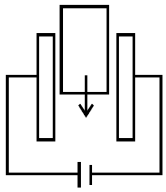
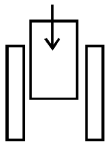
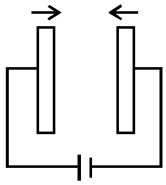
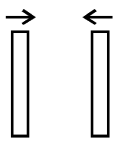
22. Two square plates with side length ℓ are arranged parallel to each other at distance d. They are charged to a potential difference V_0 and isolated from the source. A dielectric with a permittivity $\varepsilon_r = 2$ whose thickness is d and width is equal to that of the plates is drawn into the space between the plates. The length of the dielectric is greater than ℓ (figure). The resulting force F acting on the dielectric at distance $x = \ell/2$ is $\frac{\alpha C V_0^2}{\beta \ell}$ where $C = \frac{\varepsilon_0 \ell^2}{d}$. Take α and β as minimum integers.Find β-α.



SECTION-IV

Matrix Match Type (4 × 5) **3 Q. [8 M (for each entry +2(0)]**

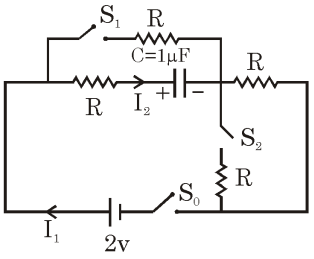
23. Column–I shows some parameters after completion of event and Column–II shows tha arrangements to which these parameters belong.

Column–I	**Column–II**
(A) Work done by external agent is negative	(P) Dielectric is being removed slowly from a charged capacitor 
(B) Electric field between plates increases	(Q) Dielectric being introduced slowly while battery is connected 
(C) Energy of capacitor decreases	(R) Dielectric is being introudced slowly in a charged capacitor 
(D) Force between plates remains constant	(S) Plates of capacitor are being slowly moved closer 
	(T) Plates of capacitor are being moved slowly in a charged capacitor 

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24. In the given figure initially all switches are open and the capacitor has charge q. At t = 0, switch S_0 is closed alongwith S_1 or S_2 or both. Column I has the entries for the actions on switches and column II has the values of currents I_1 & I_2 at t = 0. (R = 1Ω)



	Column-I		Column-II
(A)	S_0 and S_1 closed and q = 1 μC	(P)	$I_1 < 2I_2$
(B)	S_0 and S_1 closed and q = –1μC	(Q)	$I_2 = 0$
(C)	S_0, S_1 and S_2 closed and q = 1μC	(R)	$I_1 > 4I_2$
(D)	S_0, S_1 and S_2 closed and q = –1μC	(S)	$I_1 > 1A$
		(T)	$I_1 \le 1A$

25.

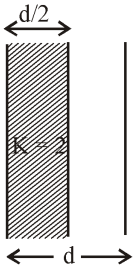
	Column-I		Column-II
(A)	Plates of an isolated, charged, parallel plate, air core capacitor are slowly pulled apart.	(P)	Electric energy stored inside capacitor increases in the process.
(B)	A dielectric is slowly inserted inside an isolated and charged parallel plate air cored capacitor to completely fill the space between plates.	(Q)	Force between the two plates of the capacitor remain unchanged.
(C)	Plates of a parallel plate capacitor connected across a battery are slowly pulled apart.	(R)	Electric field in the region between plates remain unchanged.
(D)	A dielectric slab is slowly inserted inside a parallel plate capacitor connected across a battery to completely fill the space between plates.	(S)	Total electric energy stored inside capacitor decreases in the process.
		(T)	Electric field in the region decreases.

Subjective Type **5 Q. [4 M (0)]**

26. An electron enters the region between the plates of a parallel plate capacitor at a point equidistant from either plate. The capacitor plates are 2×10^{-2} m apart & 10^{-1} m long . A potential difference of 300 volt is kept across the plates. Assuming that the initial velocity of the electron is parallel to the capacitor plates, calculate the largest value of the velocity of the electron so that it does not fly out of the capacitor at the other end.

27. Two square metallic plates of 1 m side are kept 0.01 m apart, like a parallel plate capacitor, in air in such a way that one of their edges is perpendicular, to an oil surface in a tank filled with an insulating oil. The plates are connected to a battery of e.m.f. 500 volt . The plates are then lowered vertically into the oil at a speed of 0.001 m/s. Calculate the current drawn from the battery during the process.
[di-electric constant of oil = 11, $\epsilon_0 = 8.85 \times 10^{-12}$ $C^2/N^2 m^2$]

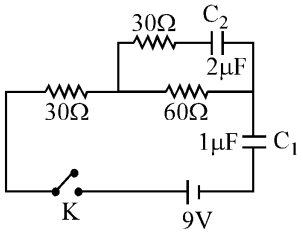
28. A certain series RC circuit is formed using a resistance R, a capacitor without dielectric having a capacitance C = 2F and a battery of emf E = 3V. The circuit is completed and it is allowed to attain the steady state. After this, at t = 0, half the thickness of the capacitor is filled with a dielectric of constant K = 2 as shown in the figure. The system is again allowed to attain a steady state. What will be the heat generated (in joule) in the circuit between t = 0 and t = ∞ ?



29. For the arrangement shown in the figure, the key is closed at t = 0. C_2 is initially uncharged while C_1 has a charge of 2μC.

(a) Find the current coming out of the battery just after switch is closed.

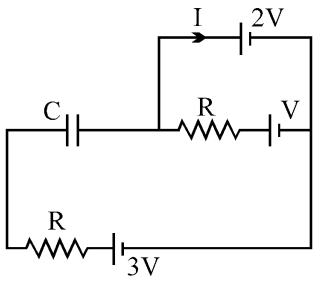
(b) Find the charge on the capacitors in the steady state condition.



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30. In the RC circuit shown, what is the variation of the current I as a function of time? The capacitor is initially uncharged and the switch is closed at time t = 0 sec.



GR # CAPACITOR **ANSWER KEY**

SECTION-I

Single Correct Answer Type **8 Q. [3 M (–1)]**

1. Ans. (B) **2. Ans. (C)** **3. Ans. (C)** **4. Ans. (B)** **5. Ans. (A)**
6. Ans. (A) **7. Ans. (A)** **8. Ans. (D)**

Multiple Correct Answer Type **7 Q. [4 M (–1)]**

9. Ans. (A,B,C) **10. Ans. (B,D)** **11. Ans. (A,C)** **12. Ans. (B,D)**
13. Ans. (A, B, C, D) **14. Ans. (A, B, C)** **15. Ans. (A, C)**

Linked Comprehension Type (2 Para × 2Q.) **[3 M (-1)]**
(Single Correct Answer Type)

16. Ans. (B) **17. Ans. (C)** **18. Ans. (A)** **19. Ans. (A)**

SECTION-II

Numerical Answer Type Question **1Q.[3M(0)]**
(upto second decimal place)

20. Ans. 4.50

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **2 Q. [4 M (0)]**

21. Ans. 42 **2. Ans. 7**

SECTION-IV

Matrix Match Type (4 × 5) **3 Q. [8 M (for each entry +2(0))]**

23. Ans. (A) → (Q, R, S, T) ; (B) → (P, S) ; (C) → (R, T) ; (D) → (P, R, T)
24. Ans. (A) → (Q,R,T) ; (B) → (P,S) ; (C) → (R,S) ; (D) → (P,S)
25. Ans. (A) → (P,Q,R) ; (B) → (Q,S,T) ; (C) → (S, T) ; (D) → (P, R)

Subjective Type **5 Q. [4 M (0)]**

26. Ans. $\frac{\sqrt{4.8}}{2\sqrt{9.1}} \times 10^8$ **m/s** **27. Ans.** 4.425×10^{-9} Ampere **28. Ans.** $\frac{3}{4}$ J

29. Ans. (a) $\frac{7}{50}$ A or $\frac{11}{50}$ A , (b) $Q_1 = 9\mu C$, $Q_2 = 0$ **30. Ans.** $\frac{V}{R}\left(e^{\frac{-t}{RC}} - 1\right)$

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